

Mounting Hardware Specifications

1. Overview

This document outlines the standard hardware types, dimensions, and torque values used when installing accessories or replacement components. The goal is to ensure consistent, safe, and repeatable installations with OEM-level reliability.

2. Hardware Types and Their Use Cases

Bolts

Used for primary structural mounting and high-load attachment points.

Common grades:

- **Grade 8.8** – General body fittings
- **Grade 10.9** – High-strength components
- **Grade 12.9** – Critical fastening or load-bearing accessories

Nuts

Paired with bolts for secure clamping force.

Types:

- **Flange nuts** – Improved load distribution
- **Nyloc nuts** – Vibration-resistant
- **Standard hex nuts** – General-purpose

Washers

Prevent surface damage and distribute load.

Types:

- **Flat washers** – For even pressure
- **Spring washers** – Added vibration resistance
- **Rubber washers** – Used where noise damping is needed

Rivets / Clips

Used for trim panels, plastic covers, and areas where screws are unsuitable.

- **Push clips** – Interior trim
- **Expansion rivets** – Wheel wells and exterior plastics

- **Metal pop rivets** – Permanent fastening points

Brackets and Plates

Provide structural support for accessories like cameras, sensors, steps, rails.

Common materials:

- Powder-coated steel
 - Aluminum
 - Reinforced polymer
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3. Standard Sizes and Dimensions

Bolts (Metric)

- **M6 x 1.0** – Light brackets, trim support
- **M8 x 1.25** – Moderate load accessories
- **M10 x 1.5** – Heavy-duty mounts
- **M12 x 1.75** – High-load structural points

Length Guidelines

- Length chosen to allow **full thread engagement** (1.5x bolt diameter minimum)
- Avoid excessive projection beyond nut or bracket

Washers

- Matched to bolt size: M6, M8, M10
- Typically 1.5–2 mm thickness depending on material

Brackets

- Minimum 3 mm thick steel plate for load-bearing accessories
 - 2 mm aluminum where weight reduction is needed
 - All edges chamfered or deburred
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4. Torque Specifications (General Guide)

Always follow the exact accessory sheet if provided. These values cover standard automotive accessory installations:

Bolt Size Grade 8.8 Grade 10.9

M6 9–11 Nm 12–14 Nm

M8 20–25 Nm 28–32 Nm

M10 40–45 Nm 48–55 Nm

M12 65–75 Nm 80–90 Nm

Rules of thumb:

- Lubricated threads reduce torque by roughly 10–15 percent
 - Never mix bolt grades on the same mount
 - Replace any bolt showing corrosion or damaged threads
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5. Quality and Safety Requirements

Material Standards

- Bolts and nuts must comply with ISO 898-1
- Washers and plates must be corrosion-resistant
- All hardware should be free from burrs, cracks, and plating defects

Installation Principles

- Always hand-start bolts before torquing
- Use cross-pattern tightening on multi-point mounts
- Verify bracket alignment before final torque
- Use thread-locker where vibration is expected (blue medium strength)

Post-Installation Checks

- Full torque verification
- No washer distortion or misalignment
- No contact with moving components
- No wiring pinched under hardware